DHRUV KOTHARI

| Materials Science Engineer | Materials Characterization | Process Engineer |

| Analytical Chemistry | Global Supply Chain & Raw materials Procurement Manager |

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PROFESSIONAL SUMMARY

Materials Science Engineer with 8+ years of experience in advanced inorganic materials synthesis, analytical characterization, process engineering, and global supply chain management serving customers in optoelectronics, semiconductor, defense optics, thin-film coating, battery/energy storage, and aerospace industries. Managed a portfolio of 10,000+ materials and SKUs and contributed to $10M+ in customer projects, including programs with Corning, Teledyne, Newport, Coherent, Pi-Kem, U.S. defense contractors like Raytheon, and leading thin-film companies. Hands on expertise with XRD, SEM/EDS, ICP-MS, FTIR, and PSD; deep knowledge of rare earth compounds, optical coating materials, specialty ceramics, sputtering targets, and high-purity metals (4N–5N). Built and led analytical and processing labs from the ground up; qualified international suppliers across India, Brazil, UK, and Europe to de-risk China-dependent supply chains.

> **★ Directly supported & supplied materials to: Corning, Teledyne, Coherent (II-VI), Materion, Kurt J. Lesker, Sigma Aldrich, Thermofisher Scientific, NASA, Tosoh SMD, Honeywell, Mitsubishi Heavy Industries, and U.S. Defense Programs**

CORE COMPETENCIES

Materials Characterization	XRD / SEM / EDS / ICP-MS	Powder Synthesis & Processing	Sputtering Target Fabrication
Thin-Film & Optical Coatings	Vacuum Induction Melting	Sintering & Densification	Rare Earth & Fluoride Compounds
Specialty Ceramics & Refractories	Battery & Electrolyte Materials	FTIR / PSD / ICP-MS / HPLC	GC-MS / UV-Vis / Flame AAS
Process Development & Scale-up	DOE & SPC / Root Cause Analysis	Impurity Control (4N–5N Purity)	Materials Qualification & QC
Global Procurement (10,000+ SKUs)	Supply Chain De-risking	Customer Technical Support	Lab Build-out & Team Leadership
Failure Analysis & Root Cause	Impurity Control & QC	Supply Chain Management	

PROFESSIONAL EXPERIENCE

Senior Materials Engineer

LTS Research Laboratories, Inc. | *Belmont, NC* | *Nov 2017 – Present*

LTS Research is a specialty materials manufacturer and supplier of high-purity inorganic compounds, rare earth materials, optical coating materials, sputtering targets, and advanced ceramics serving semiconductors, defense, optics, aerospace, and energy storage industries globally.

Scale & Business Impact

- Managed a portfolio of 10,000+ materials and SKUs spanning rare earth oxides/fluorides, optical coating compounds, sputtering targets, specialty ceramics, high-purity metals, and battery materials.
- Contributed to $10M+ in customer programs, including $1M+ engagements individually with Optoelectronics and Semiconductors Industries, multiple U.S. defense contractors, and Coherent (II-VI) — providing materials with full analytical data.
- Directly developed, qualified, and supplied materials to industry-leading customers: Corning, Teledyne , Coherent (II-VI), Materion, Kurt J. Lesker, Sigma Aldrich, Thermofisher Scientific, NASA, Tosoh SMD,

Honeywell, Mitsubishi Heavy Industries, and U.S. Defense Programs providing technical data packages, COAs, and application support.

- Built and diversified a global supplier network across India, Brazil, UK, and Europe to maintain consistent raw material supply amid China export restrictions and trade bans on rare earth and critical materials.

Materials Characterization & Analytical Chemistry

- Lead and manage all analytical operations using XRD (phase identification, crystallinity analysis), SEM/EDS (microstructure imaging, elemental mapping), ICP-MS (trace metal impurity quantification at ppb/ppt levels), PSD (particle size distribution), and FTIR (phase and compositional verification).
- Developed and validated analytical methods for high-purity inorganic compounds helping customers to achieve target film properties by confirming composition, purity (4N–5N / 99.99–99.999%), microstructure, and particle characteristics.
- Conducted materials qualification, failure analysis, and root cause investigations for production batches and customer complaints; issued detailed analytical reports and corrective action plans.
- Expanded lab analytical capabilities from XRD/SEM/PSD to include ICP-MS (sub-ppm impurity detection) and vacuum induction melting, cold press machine, tube furnace, significantly increasing in-house testing depth and speed.
- Applied ellipsometry, profilometry, and four-point probe for thin-film characterization during new optical coating material development.

Process Engineering & Materials Processing

- Designed and optimized full powder-to-part workflows: vacuum induction melting/inert muffle furnace → ball milling → blending → cold pressing → sintering → densification → machining → sputtering target finishing.
- Managed vacuum induction melting and inert atmosphere sintering furnaces for high-purity precious metals, reactive metals, refractory compounds, and advanced ceramics.
- Built in-house cold pressing and sintering production lines for metallic and ceramic sputtering targets, reducing lead times and improving consistency.
- Implemented SPC (Statistical Process Control) and DOE (Design of Experiments) to optimize synthesis parameters, reduce defects, and improve batch reproducibility.
- Scaled laboratory synthesis processes to full production, developed SOPs and process control documentation for all major product lines.

Materials Synthesis & Product Development

- Developed synthesis protocols for rare earth oxides, fluorides, carbonates, and hydroxides; optical coating materials (oxide, fluoride, nitride targets); high-purity ceramics; and specialty refractory compounds.
- Supported development of lithium-ion and sodium-ion battery cathode powders, anode materials, and solid-state electrolytes for energy storage customers.
- Created full design-to-production documentation for sputtering targets (metallic, oxide, nitride, fluoride, compound), ensuring reproducibility from lab to manufacturing scale.
- Evaluated and technically audited overseas manufacturing partners for calcination, annealing, melting, sintering, and chemical refining — ensuring all materials met specification before shipment.

Global Procurement & Supply Chain Management

- Managed global procurement of high-purity metals (precious, refractory, reactive, alkaline earth), rare earth compounds, transition metal compounds, and specialty chemicals across 10,000+ SKUs.
- Qualified and onboarded new international suppliers in India, Brazil, UK, and Europe to replace China-sourced materials impacted by export controls and trade restrictions maintaining uninterrupted supply for critical customer programs.
- Performed supplier technical due diligence, factory audits, and incoming material qualification to ensure consistent raw material quality and traceability.

Customer Technical Support & Business Development

- Served as primary technical interface for Fortune 500 and defense customers: provided guidance on material selection, sputtering target performance, thin-film deposition behavior, and coating results.
- Supported new material development from customer inquiry through prototype, analytical qualification, and commercial production — with full data packages (COA, XRD, SEM, ICP-MS reports).
- Identified and developed new material opportunities aligned with customer roadmaps in semiconductor, defense optics, aerospace, and energy storage sectors.

Quality Control Chemist

US Pharmaceuticals Inc. (Unipack) | | *Aug 2017 – Oct 2017*

- Performed HPLC assays (Empower), FTIR, viscosity, and pH analysis for raw materials and finished pharmaceutical batches.
- Ensured compliance with cGLP, cGMP, and internal SOP requirements; maintained stability testing and audit documentation.

MEMS Research Intern – Microfabrication & Thin-Film Characterization

NJIT Microelectronics Fabrication Facility | | *Jan 2016 – May 2016*

- Performed photolithography on silicon wafers (Karl Suss MA-6 aligner): spin-coating, soft-bake, UV exposure, development, and post-lithography inspection.
- Measured thin-film thickness via VEECO Dektak IIA profilometer; conducted ellipsometry and four-point probe characterization.
- Operated in ISO-class cleanroom following contamination control and ESD protocols.

Analytical Chemistry Project

NJIT | | *Sep 2016 – Dec 2016*

- LC-MS compound verification, UV-Vis quantification, Flame AAS trace metal analysis, HPLC (Shimadzu 20AC), and GC analysis for volatile organics.

Metallurgical Engineering Intern

Shree Sahaj Alloys Pvt. Ltd. | *India* | *Jun 2013 – Jul 2013*

- Supported induction furnace melting and alloy production; sand casting and pressure die casting workflows.
- Verified molten steel chemistry via optical emission spectrometry (OES); monitored heat treatment, dimensional inspection, and NDT.

EDUCATION

M.S., Materials Science and Engineering

New Jersey Institute of Technology | *2015 – 2017*

B.E., Metallurgical Engineering

Gujarat Technological University | *2011 – 2015*

TECHNICAL SKILLS

Characterization	XRD, SEM/EDS, ICP-MS, PSD, FTIR, Ellipsometry, Profilometry (VEECO Dektak), Four-Point Probe, Optical Microscopy
Analytical Chemistry	LC-MS, HPLC (Empower, Shimadzu), GC/GC-MS, UV-Vis Spectrophotometry, Flame AAS, pH, Viscosity
Processing & Fabrication	Vacuum Induction Melting, Cold Pressing, Sintering, Hot Pressing, Densification, Sputtering Target Fabrication, Powder Milling & Blending, Calcination, Annealing
Thin-Film & Cleanroom	Photolithography, Spin-Coating, Thin-Film Deposition Support, PVD/Sputtering Materials, ISO Cleanroom Protocol
Quality & Process	DOE, SPC, Root Cause Analysis (RCA), CAPA, cGMP, cGLP, SOP Development, Materials Qualification, Failure Analysis
Materials Expertise	Rare Earth Oxides/Fluorides/Compounds, Optical Coating Materials, Specialty Ceramics, Refractory Materials, Precious & Reactive Metals, Battery Cathode/Anode Powders, Solid-State Electrolytes
Operations & Leadership	Global Procurement (5,000+ SKUs), Supply Chain De-risking, Lab Management, Team Leadership & Training, Customer Technical Support
Software	SolidWorks, Microsoft Office Suite, Empower (Waters), Shimadzu Lab Solutions